ON THE APPROXIMATION OF CURVES BY LINE SEGMENTS USING DYNAMIC PROGRAMMING

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Introduction

In a recent paper [1], Stone considered the problem of determining the $2N + 2$ constants, $a_i, b_i, \quad i = 1, 2, \cdots, N + 1$ and the N points of subdivision $u_1, u_2, \cdots, u_N$ so as to minimize the function

$$F(a_1, a_2, \cdots, a_{N+1}; b_1, b_2, \cdots, b_{N+1}; u_1, u_2, \cdots, u_N) = \sum_{j=1}^{N+1} \int_{u_{j-1}}^{u_j} (g(x) - a_j - b_j x)^2 \, dx, \tag{1}$$

where $u_0 = a, \quad u_{N+1} = b$ and $a \leqq u_1 \leqq u_2 \leqq \cdots \leqq u_N \leqq b$. For the case where $g(x)$ is quadratic, one can look forward only to a simple computational algorithm for the solution. In this note we wish to show how dynamic programming provides a solution in these terms.

Basic Recurrence Relation

Let us write, for fixed a and $b \geqq a$,

$$f_N(b) = \min_{[a_i, b_i, u_i]} F(a_1, \cdots, a_N; b_1, \cdots, b_N; u_1, \cdots, u_N). \tag{1}$$

Then

$$f_1(b) = \min \left[\int_a^{u_1} (g(x) - a_1 - b_1 x)^2 \, dx + \int_{u_1}^{b} (g(x) - a_2 - b_2 x)^2 \, dx \right], \tag{2}$$

where the minimum over $-\infty < a_1, a_2, b_1, b_2 < \infty$, $a \leqq u_1 \leqq b$. This function is readily determined since we can compute the minima over the a_i and b_i and then minimize over u_1 by means of a discrete search.

It is easy to see that for $N \geqq 2$ we have

$$f_N(b) = \min_{0 \leqq u_N \leqq b} \left[\min_{[a_N, b_N]} \int_{u_N}^{b} (g(x) - a_N - b_N x)^2 \, dx + f_{N-1}(u_N) \right]. \tag{3}$$

This is a particular application of the principle of optimality [2, 3]. Since the minimum over a_N, b_N can be calculated explicitly, we can write (3) in the form:

$$f_N(b) = \min_{a \leqq u_N \leqq b} [h(u_N, b) + f_{N-1}(u_N)] \tag{4}$$

A computational solution along these lines requires a few seconds per stage, where N is the number of stages; see [3].

Extensions

It requires very little additional effort to approximate to $g(x)$ by quadratic polynomials, or by polynomials of any fixed degree. Similarly, we can compute the minimum of

$$\sum_{j=1}^{N+1} \int_{u_{j-1}}^{u_j} (g(x) - h(x, a_j, b_j))^2 \, dx, \tag{1}$$

provided that we know how to minimize the function

$$\int_{u_N}^{b} (g(x) - h(x, a_N, b_N))^2 \, dx \tag{2}$$

over a_N and b_N, or the minimum of

$$\sum_{j=1}^{N+1} \max_{u_{j-1} \leqq x \leqq u_j} |g(x) - h(x, a_j, b_j)|. \tag{3}$$

REFERENCES

1. STONE, H. Approximation of curves by line segments. *Math. Comput. 15* (1961), 40–47.
2. BELLMAN, R. *Dynamic Programming.* Princeton University Press, Princeton, N. J., 1957.
3. BELLMAN, R., AND DREYFUS, S. *Computational Aspects of Dynamic Programming.* Princeton University Press, Princeton, N. J.; to appear.

Letters (Continued from page 253)

Irons' Procedure DIAGRAM

Dear Editor:

I would like to make some comments on the excellent and stimulating article, "A Syntax Directed Compiler," by Edgar T. Irons in the *Comm. ACM* issue of Jan. 1961, a preprint of which has been used here in the design of an ALGOL 60 translator for GIER, a new Danish computer.

I think that the ALGOL formulation of the procedure DIAGRAM given at the end of the article should have the following corrections:

(1) line 17 *should begin* **if** i ≠ 0.....
(2) line 19 *should begin* **if** STAB [i + 1] =.....
(3) line 27 *should begin* **if** i ≠ 0.....

(4) OTCEL should be assigned some "unconfusing" value before the statement labelled START is obeyed, as otherwise it will be undefined in the statement labelled FOUND in the case when STAB [i] = INPUT [j] and STAB [i + 1] ≠ LEFTBRACE.

It must be common to many languages that could be translated with the aid of this procedure that SUCCR [INPUT [j], INPUT [j]] is always *false* and so lines 10–15 of DIAGRAM can be greatly simplified.

B. H. MAYOH
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